

Unrestricted pro-modularity at the prime two

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Abstract

Every continuous, odd, absolutely irreducible two-dimensional 2-adic representation of $G_{\mathbb{Q}}$ that is unramified outside finitely many finite primes occurs in the full completed Hecke algebra at some odd tame level. The level may contain auxiliary tame primes, and scalar and reducible residual representations are included. This is a completed-Hecke occurrence result, with no de Rham hypothesis; it does not assert classical modularity.

1 The full Hecke algebra and the theorem

Congruences between modular forms of different weights assemble their Hecke eigenvalues into a single p -adic algebra. Its points need not come from classical forms of one weight. The resulting modularity question asks whether a Galois representation occurs in this larger spectrum, even when no Hodge-theoretic condition is imposed at p . We prove this occurrence statement for every odd, absolutely irreducible two-dimensional representation over $\mathbb{Q}$ at $p = 2$.

Fix an odd positive integer N . For $i \geq 1$, let $M_i(\Gamma_1(N))$ denote the space of all complex holomorphic modular forms of weight i , including Eisenstein forms. Let $T_{\leq k}^{(2)}(N)$ be the $\mathbb{Z}$ -algebra acting on $\bigoplus_{i=1}^k M_i(\Gamma_1(N))$ generated by T_ℓ and ℓS_ℓ for primes $\ell \nmid 2N$. On weight i , the operator S_ℓ is $\ell^{i-2}\langle \ell \rangle$. Set

$$\mathbb{T}_2(N) = \varprojlim_k (\mathbb{Z}_2 \otimes_{\mathbb{Z}} T_{\leq k}^{(2)}(N)). \quad (1)$$

Each finite-stage algebra has its 2-adic topology, and $\mathbb{T}_2(N)$ has the resulting inverse-limit topology. We fix the Frobenius convention in which a classical eigenform has characteristic polynomial

$$Z^2 - T_\ell Z + \ell S_\ell \quad (2)$$

at a good prime, as in [18, Equation (2.2)].

Theorem 1.1. *Let $E/\mathbb{Q}_2$ be finite, with ring of integers $\mathcal{O}_E$, and let $r : G_{\mathbb{Q}} \rightarrow \mathrm{GL}_2(E)$ be continuous and absolutely irreducible. Suppose that r is unramified outside finitely many finite primes and is odd: $\det r(c) = -1$ for complex conjugation c . Then there exist an odd positive integer N , divisible by every odd prime where r ramifies, and a continuous $\mathbb{Z}_2$ -algebra homomorphism*

$$\lambda : \mathbb{T}_2(N) \rightarrow \mathcal{O}_E$$

such that, for every prime $\ell \nmid 2N$,

$$\lambda(T_\ell) = \mathrm{tr} r(\mathrm{Frob}_\ell), \quad \lambda(\ell S_\ell) = \det r(\mathrm{Frob}_\ell).$$

We call this conclusion *pro-modularity*. No condition on the semisimplified reduction of r is imposed. In particular, the theorem includes scalar residual representations. The compact image of r preserves an $\mathcal{O}_E$ -lattice, so its traces and determinants are integral.

Context and main ideas

The Fontaine–Mazur conjecture relates Galois representations satisfying geometric local conditions to algebraic geometry. In its odd, two-dimensional regular form over $\mathbb{Q}$, the expected conclusion is classical cuspidal modularity up to a Tate twist [12, 19]. Removing the local condition leads to a different question: occurrence of the representation in a completed Hecke algebra. Emerton records this expectation for the full varying-weight algebra in [11, Conjecture 2.12]. That formulation prescribes the allowed tame primes. Theorem 1.1 proves dyadic occurrence for absolutely irreducible representations while allowing auxiliary primes in the tame level.

Skinner and Wiles developed pro-modularity arguments through ordinary families in the residually reducible setting [22, Introduction and Section 4.1]. Emerton’s local–global compatibility work separates pro-modularity from the production of classical vectors [10, Section 1.2]; the pro-modularity theorem recorded there assumes $p > 2$, residual irreducibility, and local residual restrictions. Pan extends the family method to the residually reducible nonordinary setting at odd primes: ordinary loci supply modular points, and patching at one-dimensional primes propagates Hecke support [19, Introduction and Theorem 4.1.7]. His classical modularity theorems retain regular de Rham hypotheses. These precedents suggest placing an unrestricted representation in a family and finding a useful geometric point elsewhere in that family.

The local representation theory used here comes from Colmez’s p -adic Langlands correspondence [7], in the all-prime form of Colmez, Dospinescu, and Paškūnas [8, Theorem 1.1]. Paškūnas and Tung supply the integral block finiteness and Cayley–Hamilton descriptions needed at 2, including scalar residual blocks [20, Theorems 1.2–1.4 and Section 4.1]. Their results control local parameters but do not by themselves place a global representation on Hecke support.

After a continuous cyclotomic twist, we place the original representation in a fixed-determinant trace family D of dimension at least three. The curve and connectedness arguments of [17] put this family on Hecke support after solvable totally real base change, as developed in Section 2. The main task is then to find a regular de Rham point on D , from which Hecke support can be transferred to the whole family over $\mathbb{Q}$.

Two constructions proved here produce that point. We choose the base-change field to split completely at 2, so that all dyadic restrictions of the family have the same local Galois parameter. Section 3 passes through the local block equivalences to construct an object for one copy of $\mathrm{GL}_2(\mathbb{Q}_2)$ from an admissible representation of the product of dyadic groups. Equality of the parameters ensures that a failure of admissibility would persist on a characteristic-two curve in every factor, contradicting the admissibility of the product representation. This comparison includes scalar residual blocks. Section 4 then uses a chain of prime specializations in D to force maximal growth, and hence positive Iwasawa rank, for the one-factor object. Polynomial approximation detects a locally algebraic depth-zero supercuspidal type, yielding a regular de Rham point x . The regular Fontaine–Mazur theorem [17, Theorem 1.1] makes x classical up to Tate twist, and Section 5 puts its classical packet and continuous cyclotomic twists in the precise algebra (1).

The final passage uses the dimension theorem of [18, Theorem 1.1] and the adjoint Selmer vanishing theorem of Newton and Thorne [16, Theorem 5.4]. At x , the global deformation problem with varying determinant has tangent dimension at most three, while a dimension-four integral Hecke component has local dimension three. Equality makes the ambient local ring regular and forces its Hecke kernel to vanish there. Since D is a domain through x , the same kernel vanishes on all of D . Section 6 carries out this comparison and recovers the original coefficient point. The Hodge-theoretic condition is needed only at the auxiliary point.

Coefficient and deformation conventions

We allow finite extensions of coefficient fields during the proof. Write $\mathcal{O}$ for the current ring of integers, ϖ for a uniformizer, and k for the residue field. A *coefficient point* of a complete local $\mathcal{O}$ -algebra is a continuous map to the ring of integers of a finite coefficient extension. Its kernel is a characteristic-zero prime; the corresponding integral quotient has dimension one. A domain over $\mathcal{O}$ is *horizontal* if ϖ is nonzero in it.

At residue characteristic two, a pseudorepresentation means a two-dimensional determinant law, with both trace and determinant data, in the sense of [5]. Thus no division by 2 is implicit in an integral deformation ring. All global deformation problems have a fixed finite ramification set containing 2 and allow ramification at infinity. Trace images and support containments are closed images and set-theoretic containments; the domains used for the family are reduced.

For the local categories and completed definite forms we use the normalization of [17, Section 2]. If the fixed global determinant is χ , the central character is $\psi = \chi\varepsilon$, where ε is the cyclotomic character. The local block parameter is the Galois parameter twisted by ε , of determinant $\psi\varepsilon = \chi\varepsilon^2$. Group duals have the contragredient action; commuting coefficient and Hecke operators act by transposition. The conversion to the Frobenius identity (2) is made explicitly when passing to classical Hecke points.

2 A trace family with potential automorphic support

We first place the given representation in a sufficiently large fixed-determinant family. We then put that entire family on completed Hecke support after a totally real base change. The representation itself need not have an algebraic weight. The distinction between its residual representation and an auxiliary representation supplying the determinant will be essential in the support argument. For a matrix representation, *not virtually solvable* means that the identity component of its algebraic monodromy group is not solvable. This property is unchanged on restriction to a finite index subgroup.

2.1 Normalizing the determinant

We will use the following twist property, proved in Proposition 5.2: at an odd tame level $N \geq 5$, twisting a Hecke coefficient point by any continuous character $\eta : \mathbb{Z}_2^\times \rightarrow \mathcal{O}'^\times$ gives another point of the same Hecke algebra. At a good prime ℓ , this multiplies the trace by $\eta(\varepsilon(\text{Frob}_\ell))$ and the determinant by its square. Thus it suffices to prove the theorem after any such twist of r , and to undo the twist at the end.

Lemma 2.1. *After a finite coefficient extension and a twist through ε , one may arrange*

$$\det r = \chi = \delta\varepsilon^w,$$

where δ has finite order and is unramified at 2. The integer w may be chosen arbitrarily large in one parity class. It may in particular be chosen so that χ is the determinant of a regular cuspidal modular representation that is not virtually solvable. Oddness is preserved. The character $\psi = \chi\varepsilon$, viewed on $\mathbb{Z}_2^\times$ by local reciprocity, is $z \mapsto z^{w+1}$.

Proof. Global class field theory separates the dyadic part of $\det r$ from its odd conductor part:

$$\det r = \delta \alpha \circ \varepsilon, \quad \alpha : \mathbb{Z}_2^\times \longrightarrow \mathcal{O}^\times.$$

Here δ has finite order and odd conductor. Indeed, at an odd prime the pro- ℓ group of principal units has finite image in $\mathcal{O}^\times$, whose open subgroup is pro-2; only finitely many primes occur. The remaining unramified global character is trivial by the class field theory of $\mathbb{Q}$.

Choose w with $(-1)^w = \alpha(-1)$. Write $\mathbb{Z}_2^\times = \{\pm 1\} \times 5^{\mathbb{Z}_2}$. The value $\alpha(5)$ reduces to one, since a pro-2 group has trivial image in $k^\times$. After finite coefficient extension choose

$$u^2 = 5^w \alpha(5)^{-1}.$$

The element u is integral and reduces to one. Its powers u^{2^n} tend to one, so $a \mapsto u^a$ defines a continuous character of $\mathbb{Z}_2$. Define $\beta(-1) = 1$ and $\beta(5) = u$. Then $\alpha(z)\beta(z)^2 = z^w$ for every $z \in \mathbb{Z}_2^\times$. Replacing r by $r \otimes (\beta \circ \varepsilon)$ gives the asserted determinant. Since $\beta(-1) = 1$, complex conjugation and oddness are unchanged. Local reciprocity identifies ε on dyadic units with the identity character; δ is unramified there.

For completeness, the auxiliary modular representation can be chosen without imposing the residual representation of r . Oddness gives $\delta(-1) = (-1)^{w+1}$. Fix an odd level divisible by the conductor of δ , and enlarge it to a fine level. The dimension formula [6, Section III, Théorème 1] for cusp forms with this character grows linearly with the weight $w + 1$ along the indicated parity class. The CM contribution at this fixed level is bounded independently of the weight. To see the latter assertion, a CM eigenform comes from a Hecke character of an imaginary quadratic field K ; its level is $|\text{disc } K| \text{Norm}(\mathfrak{f})$, where $\mathfrak{f}$ is the character conductor [21, Theorem (3.4), Corollary (3.5), and Theorem (4.5)]. Only finitely many pairs $(K, \mathfrak{f})$ occur at the fixed level. For a fixed infinity type the characters with conductor dividing a fixed ideal form, when nonempty, a torsor under a finite ray class character group. Oldform multiplicities are bounded by the fixed level as well. Thus a sufficiently large w gives a non-CM cuspidal eigenform of determinant $\delta \varepsilon^w$. Its representation is regular and not virtually solvable. Indeed its restriction to every finite index subgroup is semisimple; if its connected algebraic monodromy were solvable, its unipotent radical would act trivially, so that connected group would be a torus. An open subgroup would then have abelian image, which Ribet's CM criterion excludes [21, Propositions (4.2) and (4.4)]. $\square$

We henceforth use the normalized r and enlarge the finite coefficient ring $\mathcal{O}$ whenever necessary. Every enlargement is finite. When a domain splits after coefficient extension, choose a component through the specified coefficient point. Finite flat scalar extension makes its minimal prime contract to the old minimal prime, so its dimension is unchanged. Reduced support containments also persist under this operation. Thus the original coefficient point and the dimension bounds below are retained.

2.2 The dimension of the trace family

Lemma 2.2. *Fix a finite ramification set S containing 2, infinity, and the ramified primes of the normalized r . There is a reduced horizontal framed deformation component B of determinant χ through r . Its completed trace image D is a complete local $\mathcal{O}$ -domain, contains the coefficient point of r , and satisfies*

$$\dim B \geq 6, \quad \dim D \geq 3.$$

Moreover, B has a coefficient point whose representation is not virtually solvable. Consequently the restriction of this family to any finite extension of $\mathbb{Q}$ is generically absolutely irreducible.

Proof. Choose a stable lattice of r and let $R_{r,S}^{\square,\chi}$ be its unrestricted framed deformation ring with determinant χ . The residue here is the actual lattice reduction; no semisimplicity or irreducibility

is imposed on it. These rings and their maps to fixed-determinant pseudodeformation rings exist under Mazur's finiteness condition, which holds for $G_{\mathbb{Q},S}$ [17, Proposition 3.1].

Complete at the characteristic-zero point of r . Its real local framed deformation ring is smooth of dimension two: an odd involution has eigenvalues 1, -1 , and its conjugacy orbit has stabilizer the two-dimensional diagonal torus. For the global ring over this real local ring, the relative relation estimate of [17, Proposition 3.6], with no imposed dyadic condition, is

$$g - r_{\text{rel}} \geq 3 - h^0((\text{ad}^0 r)^\vee(1)) = 3.$$

The equality follows from absolute irreducibility. In characteristic zero the trace pairing identifies $\text{ad}^0 r$ with its dual; a nonzero invariant in its cyclotomic twist would give a nonzero map $r \rightarrow r \otimes \varepsilon$. Such a map is an isomorphism, whereas its determinants would require $\varepsilon^2 = 1$. Every component of the completed local ring therefore has dimension at least $2 + 3 = 5$. The dimension comparison at a coefficient point in [17, Lemma 3.4] adds one on returning to a horizontal integral component. Choose such a component through r and give it the reduced structure; its ring B has dimension at least six.

Let $R_S^{\text{ps},\chi}$ be the global pseudodeformation ring of the semisimplified lattice reduction, and set

$$D = R_S^{\text{ps},\chi} / \ker(R_S^{\text{ps},\chi} \rightarrow B).$$

Write $T(g) \in D$ for its universal trace at g . The ring D is a horizontal complete local domain. Its image in B is closed: complete local rings with finite residue field are compact, and their continuous images in Hausdorff rings are closed. The generic representation on B is absolutely irreducible, since the irreducible locus contains r . Forgetting its single frame loses at most three dimensions, by [17, Lemma 3.2]. Hence $\dim D \geq \dim B - 3 \geq 3$.

We justify the last assertion by bounding the trace loci of the virtually solvable representations. The reducible locus is closed and proper on D , because it does not contain r . An absolutely irreducible virtually solvable two-dimensional representation is induced from a character of a quadratic field, or has finite projective image. This follows by considering the connected solvable algebraic subgroup: a noncentral torus has two eigenlines permuted by the whole group, while a central connected subgroup gives finite projective image. A nontrivial connected unipotent subgroup would have a unique invariant line and force reducibility.

Only finitely many quadratic fields K can occur. The corresponding quadratic self-twist is unramified outside S , so this is Hermite–Minkowski with bounded local degree. For each K , the maximal abelian pro-2 quotient of $G_{K,S}$ has rank at most two: class field theory bounds its free part by the dyadic unit groups, of total rank $[K : \mathbb{Q}] = 2$; class groups and non-dyadic units contribute only finite groups. The invariant part under the quadratic involution has rank at least one, from the cyclotomic extension. If the inducing character is θ , fixing the determinant fixes $\theta\theta^s$ on G_K . Thus at most one free character parameter remains. More explicitly, fixing the character on the image of $1 + s$ fixes a sublattice of rank at least one; the finite index in its saturation contributes an integral, finite extension, not an extra parameter. The resulting complete character rings have integral dimension at most two. There are only finitely many possible residual characters, since their unordered pair is prescribed by $\bar{r}|_{G_K}$.

Their images in the trace space are closed of dimension at most two. Indeed, each generating character value satisfies

$$X^2 - T(h)X + \chi(h) = 0 \quad (h \in G_K).$$

Finitely many generators of the abelian character group, followed by complete Nakayama, make each character ring finite over its trace image. This is the finite-character argument of [17, Proposition 3.1(3)].

For finite projective image, the classification of finite subgroups of PGL_2 leaves cyclic and dihedral groups, already covered by reducibility and induction, and the three exceptional groups A_4, S_4, A_5 . Their exponents divide $M = 60$. For any matrix with projective order dividing M , the ratio ζ of its eigenvalues satisfies $\zeta^M = 1$, and

$$\frac{T(g)^2}{\chi(g)} = 2 + \zeta + \zeta^{-1}.$$

Consequently $T(g)$ is a root of the monic polynomial

$$\prod_{\zeta^M=1} (X^2 - \chi(g)(2 + \zeta + \zeta^{-1})) \in \mathcal{O}[X].$$

Its coefficients belong to $\mathcal{O}$ because the product over roots of unity is Galois invariant and integral. The fixed-determinant pseudo ring is topologically generated by finitely many traces: traces generate it by the determinant-of-a-sum identity, and finitely many suffice by Nakayama on its cotangent space. Imposing these monic equations on such generators gives a finite $\mathcal{O}$ -algebra. Thus the exceptional finite-projective trace locus has integral dimension at most one.

The union of these finitely many closed loci cannot contain the generic point of D , whose dimension is at least three. Pull it back to B and use coefficient-point density [17, Lemma 3.3(2)] to obtain a point outside it. Restriction to any finite index subgroup preserves the nonsolvable identity component of its monodromy, so that point remains absolutely irreducible. The same is then true generically on the restricted family and on every component containing that family. $\square$

2.3 Potential support with independently prescribed residue

A closed trace locus over a totally real field is *pro-modular* if it lies in the spectrum of a residual factor of the completed Hecke algebra of definite quaternionic forms, with the prescribed central character and one tame level. It is *potentially pro-modular* if this holds after a finite totally real solvable extension, split completely above 2. All containments here use the reduced closed trace images. This is the support convention of [17, Section 2.3].

The potential pro-modularity theorem in [17, Theorem 7.1] assumes a regular de Rham target. Our family need not contain such a target at this stage. We therefore use its curve and connectedness arguments with the determinant and residue specified separately.

Proposition 2.3. *Let B be the family in Lemma 2.2, with the normalized determinant of Lemma 2.1. There is a solvable totally real Galois extension $F/\mathbb{Q}$, of even degree and completely split at 2, such that the closed trace image of $B|_{G_F}$ is pro-modular at one fixed tame level and central character $\psi|_{G_F}$. This assertion includes scalar and reducible semisimple residual representations.*

Proof. The determinant and the residual representation enter this argument separately. Lemma 2.1 supplies a determinant arising from a regular modular representation; the modular seed below will instead lift the residual representation of B . We connect a characteristic-two curve on the seed's Hecke support to one in an unrestricted locus containing the restricted family. The constraints are used only to select and connect these curves. Once the second curve has potential Hecke support, localized propagation puts the entire unrestricted locus on that support.

We use the following precise propagation input. The localized propagation theorem [17, Theorem 5.1] takes a totally real solvable field of even degree, completely split at 2, a finite allowed set, a determinant of the form χ supplied by a regular target, and, separately, a semisimple residual representation of determinant $\bar{\chi}$. If a characteristic-two curve in this pseudodeformation space

is potentially pro-modular, has non-virtually-solvable generic representation, and has finite local images at the allowed non-dyadic places, then every irreducible closed locus containing that curve is potentially pro-modular.

Choose a regular cuspidal modular lift of this semisimple residue, using [17, Lemma 7.2]. That lemma includes character sums and scalar residue. Enlarge the initial allowed set to contain its tame primes and those of B , χ , and the auxiliary regular representation, before making any field choices. Field preparation [17, Proposition 6.1] provides abelian totally real 2-extensions

$$\mathbb{Q} \subset A \subset F_0$$

in which 2 splits completely. Put $d_A = [A : \mathbb{Q}]$ and $d = [F_0 : \mathbb{Q}]$. The numbers u_A, u_0 of allowed non-dyadic primes are bounded by a constant U while both degrees grow. Absolutely irreducible residue remains absolutely irreducible. In the reducible case write $\bar{r} = \bar{\chi}_1 \oplus \bar{\chi}_2$ and $\alpha = \bar{\chi}_1/\bar{\chi}_2$. Distinct characters remain distinct, and restriction from A to F_0 is injective on the residual extension groups for α and α^{-1} .

When the global residue is reducible and its dyadic characters coincide, the connectedness argument will keep track of chosen local invariant lines at selected dyadic places. These places are the marks. Field preparation lets us make their number large while keeping their proportion among all dyadic places small.

There are no marks if the residue is absolutely irreducible or $\alpha|_{G_{\mathbb{Q}_2}} \neq 1$. Otherwise field preparation specifies one dyadic place v_0 of A , and we mark all its $m = [F_0 : A]$ extensions to F_0 . Thus $m/d = 1/d_A$; put $m = 0$ in the unmarked case. We choose the degrees so that

$$d > u_0 + 4, \quad m > u_0 + 4 \text{ if marked}, \quad 2d - 2m - 4 - u_0 > d. \quad (3)$$

These are the degree conditions in [17, Section 7.2]. Any further fixed lower bound on d_A is compatible with them.

We now construct two large loci over A . Solvable base change and Jacquet–Langlands transfer the modular seed to definite forms. Regularity keeps this base change cuspidal [17, Lemma 4.6]. Its central character and the prescribed ψ agree modulo ϖ . At a sufficiently deep tame level, the integral function description and exact reduction of completed forms therefore give a nonzero residual summand with central character precisely ψ ; this is the seed construction in [17, Section 7.4]. A component C_{seed} of its Hecke support has

$$\dim C_{\text{seed}} \geq 1 + 2d_A$$

by [17, Proposition 4.4].

Independently choose an unrestricted fixed-determinant framed component containing the restricted family $B|_{G_A}$, by taking a minimal prime below its defining prime. Its generic representation is absolutely irreducible by Lemma 2.2. The relative presentation over the real local factors gives the component dimension at least $1 + 2d_A + 3 - h^0((\text{ad}^0)^\vee(1))$; the invariant term is at most three. Forgetting the frame loses at most three more dimensions. Thus its trace image C_{fam} , which contains the restricted trace family, has

$$\dim C_{\text{fam}} \geq 2d_A - 2$$

by [17, Proposition 3.6, Lemma 3.2, and Section 7.4].

In both loci impose $\varpi = 0$ and the residual trace of one Frobenius element at each allowed non-dyadic place. In the marked case impose also the constant residual local pseudorepresentation at v_0 . The latter costs at most κ equations, where κ is the number of generators of the closed-point

ideal in one fixed dyadic pseudo ring; it is independent of d_A . Put $\kappa = 0$ when unmarked. Taking $d_A > U + \kappa + 3$, the cut in C_{fam} has dimension at least

$$2d_A - 3 - u_A - \kappa > d_A;$$

the cut in C_{seed} has at least as large a lower bound. Over the abelian field A , the characteristic-two reducible locus has dimension at most one, the dihedral locus at most d_A , and finite-projective-image curves are constant [17, Lemma 3.8]. Curve avoidance [17, Lemma 3.3(1)] therefore gives a non-virtually-solvable characteristic-two curve in each cut. The Frobenius conditions make their allowed non-dyadic local images finite by [17, Lemma 3.9]. Denote these curves by C_s and C_f .

It remains to transfer support from C_s to C_f . Over F_0 , consider the characteristic-two framed representation scheme of the fixed semisimple residue, adjoining an invariant projective line at every mark. A closed residual point is called *transverse* if none of these marked lines is globally invariant; without marks this is an empty condition. We check that both curves enter transverse residual charts. Without marks, any stable lattice works. With marks, the local pseudo at v_0 is constant and scalar after a character twist. The curve representation thus has a local invariant line, after a finite extension of its field. For a nonzero vector v on that line, over the normalized curve ring $J = k'[[t]]$ form

$$\mathcal{V} = \sum_{g \in G_A} J\rho(g)v.$$

Compactness bounds this module in a lattice, and absolute irreducibility makes it a full stable lattice. The vector v is primitive: otherwise every translate would belong to $t\mathcal{V}$, and Nakayama would give $\mathcal{V} = 0$. Its reduced orbit spans $\mathcal{V}/t\mathcal{V}$, so its reduction belongs to no global invariant line.

This avoidance survives restriction to F_0 . A nonsplit extension of distinct residual characters remains nonsplit by the restriction injectivity, while a split representation retains its two distinct character lines. For globally scalar semisimplification, the orbit condition rules out a split scalar reduction; its nonzero additive extension remains nonzero by the same injectivity. These statements hold after any residual field extension. Finally, F_0/A is Galois, so transporting the local line to all conjugate marks preserves avoidance of every global invariant line. Both curves therefore pass through transverse residual points, including in the scalar residual case.

We recall what the two connectedness inputs prove, to make the support transfer explicit. At a transverse point, [17, Lemma 7.3] gives components of the completed chart of dimension at least $2d - m$, with successive intersections in a connecting chain of dimension at least $2d - m - 1$. The reducible locus has strictly smaller dimension. For an intersection component of maximal dimension, forgetting its frame and marked lines loses at most $3 + m$ dimensions, and forgetting the characters on those lines is finite. Its trace image therefore has dimension at least $2d - 2m - 4$. The u_0 Frobenius trace conditions leave dimension greater than d by (3), so the same bad-locus avoidance supplies a non-virtually-solvable curve with finite bad local images. Localized propagation carries support across each such intersection. This is [17, Lemma 7.4].

The transverse residual fiber is geometrically connected by [17, Lemma 7.5]. In particular this includes scalar residue: after fixing the unique global invariant line, the nonzero additive extension classes and the transverse marked lines give a connected parameter space. The finite graph of components meeting this fiber therefore joins the seed chart to the family chart. Starting from C_s , which is pro-modular, the preceding chart propagation makes $C_f|_{G_{F_0}}$ potentially pro-modular. Hence C_f over A is potentially pro-modular. Apply localized propagation now over A to the *unrestricted* locus C_{fam} containing C_f . The entire locus, and therefore the prescribed family, acquires Hecke support. Marks have served only to connect curves in characteristic two; they impose no condition on B .

Only finitely many components, charts, and propagation steps occur. Taking the common Galois compositum of the resulting fields and a common deeper tame level, as in [17, Lemma 4.6 and Section 7.4], gives one extension $F/\mathbb{Q}$ with all the stated properties. $\square$

2.4 The diagonal local parameter and completed forms

Write D for the trace domain of Lemma 2.2 and choose F from Proposition 2.3. Let $\mathbb{T}$ be the residual completed Hecke algebra over F just obtained. Support containment gives a continuous map $\mathbb{T} \rightarrow D$: its image is the closed algebra of traces restricted to G_F . Indeed the Hecke kernel vanishes in this reduced domain, by the definition of closed support.

Let R be the local pseudodeformation ring of the residual dyadic parameter, with determinant $\chi|_{G_{\mathbb{Q}_2}} \varepsilon^2$. This is *block normalization*: we twist each Galois parameter by ε , and the central character is $\psi = \chi\varepsilon$. Because 2 splits completely in F , each of the $b = [F : \mathbb{Q}]$ dyadic restrictions of the global family has this same parameter as a D -valued determinant. The identifications use conjugate decomposition groups; traces are unchanged by the conjugations.

For the corresponding compact dual block of $G = \mathrm{GL}_2(\mathbb{Q}_2)$ with this central character, let P be the direct sum of one projective envelope of the dual of each simple object, and put $\mathcal{E} = \mathrm{End}(P)$. Coefficients are enlarged so that the block is absolutely split. The projective-generator functor $\mathrm{Hom}(P, -)$ takes values in right $\mathcal{E}$ -modules; its inverse is completed tensor product with P . We use the exact compact block equivalence here, and recall its finer local properties in Section 3.

The next proposition puts the trace family and these local actions on one coefficient module. Its finiteness over D and R will let us compare specialization with admissibility.

Proposition 2.4. *For the normalized representation r , the preceding constructions give a horizontal complete local domain D of dimension at least three containing its coefficient point, and continuous maps*

$$\mathbb{T} \longrightarrow D, \quad R \longrightarrow D$$

that make D finite over both source rings. There is a right $\mathcal{E}^{\widehat{\otimes} b}$ -module L_0 , finite and faithful over $\mathbb{T}$, whose inverse under the product block equivalence is the residual summand of compact definite forms. The module

$$L = (L_0 \otimes_{\mathbb{T}} D) / (D\text{-torsion})$$

is nonzero, finite and torsion-free over D . All b central R -actions on L are the same map $R \rightarrow D$. Its inverse under the product block equivalence is finite over the completed group algebra of a sufficiently small product of determinant-one compact subgroups.

Proof. Only finiteness and the module assertions remain to be proved. Let D_F be the closed restricted trace image in D . If $n = [F : \mathbb{Q}]$, then $g^n \in G_F$ for every $g \in G_{\mathbb{Q}}$. The trace-of-powers recurrence

$$P_0(X, a) = 2, \quad P_1(X, a) = X, \quad P_j(X, a) = X P_{j-1}(X, a) - a P_{j-2}(X, a)$$

gives a monic polynomial $P_n(X, a)$ of degree n , with

$$P_n(T(g), \chi(g)) = T(g^n) \in D_F.$$

Choose finitely many traces topologically generating D over $\mathcal{O}$. Each is integral over D_F , so the subalgebra they generate is finite over D_F . It is compact and hence closed in D ; since it also contains a dense subalgebra, it equals D . Thus D is finite over D_F , and therefore over $\mathbb{T}$.

The local-global block comparison [17, Proposition 4.2] makes $\mathbb{T}$ finite over $R^{\widehat{\otimes} b}$, with its actual Galois restriction action in block normalization. On D this action factors through

$$R^{\widehat{\otimes} b} \longrightarrow R, \quad a_1 \otimes \cdots \otimes a_b \longmapsto a_1 \cdots a_b,$$

because all dyadic parameters agree. Transitivity of finiteness then proves that D is finite over the single ring R .

Let M be the compact definite-forms summand. The same proposition gives

$$L_0 = \mathrm{Hom}_{G^b, \mathrm{cts}}(P^{\widehat{\otimes} b}, M),$$

finite and faithful over $\mathbb{T}$, with the asserted compatible product action. The compact module M is admissible, equivalently finite over a sufficiently deep determinant-one completed group algebra, by [17, Proposition 4.1].

At the generic point of D , the base change of L_0 is nonzero. Indeed, if $\mathfrak{q}$ is its contraction to $\mathbb{T}$, faithfulness and finiteness give $\mathrm{Supp}_{\mathbb{T}} L_0 = \mathrm{Spec} \mathbb{T}$; Nakayama then gives $L_0 \otimes_{\mathbb{T}} \kappa(\mathfrak{q}) \neq 0$. Extending this residue field to $\mathrm{Frac}(D)$ preserves nonvanishing. Quotienting by D -torsion consequently leaves a nonzero finite D -module L . The torsion submodule is stable under every $\mathcal{E}$ -action because those actions commute with D . Since D is horizontal, L is also $\mathcal{O}$ -torsion-free.

Finally, finiteness of D over $\mathbb{T}$ makes $L_0 \otimes_{\mathbb{T}} D$, and then L , a quotient of a finite direct sum of copies of L_0 as product-block modules. These are maps of compact modules with closed images. Exactness of the product block equivalence makes the inverse object of L a quotient of the corresponding finite sum of copies of M . It is therefore admissible as asserted. This supplies product admissibility; passing to a single factor is the separate argument of the next section. $\square$

3 Admissibility from diagonal local parameters

Proposition 2.4 gives an admissible compact object for G^b , where $G = \mathrm{GL}_2(\mathbb{Q}_2)$ and $b = [F : \mathbb{Q}]$. We need an admissible object for one copy of G , still carrying the whole trace family D . The equality of the b local Galois parameters is the additional information that makes this possible.

Retain the local block generator P , its endomorphism algebra $\mathcal{E} = \mathrm{End}(P)$, and its fixed-determinant pseudodeformation ring R from Proposition 2.4. Thus the central character is $\psi = \chi\varepsilon$, the Galois determinant in block normalization is $\psi\varepsilon$, and the compact category uses the dual central action. Write

$$\mathcal{E}^{(b)} = \widehat{\bigotimes}_{i=1}^b \mathcal{E}, \quad P^{(b)} = \widehat{\bigotimes}_{i=1}^b P.$$

All completed tensor products in this section are in the category of pseudocompact modules. Coinvariants mean quotients by the closed submodule generated by the indicated relations.

Proposition 3.1 (One-factor admissibility). *Let D be a complete noetherian local $\mathcal{O}$ -domain in which $\varpi \neq 0$, and suppose that $R \rightarrow D$ is a finite continuous local map. Let $L \neq 0$ be a finite torsion-free D -module with a commuting continuous right $\mathcal{E}^{(b)}$ -action. Suppose that every copy of R acts through the same map $R \rightarrow D$, and that*

$$M = L \widehat{\otimes}_{\mathcal{E}^{(b)}} P^{(b)}$$

is admissible in the compact category. Then, using the first $\mathcal{E}$ -action,

$$V = L \widehat{\otimes}_{\mathcal{E}} P$$

is nonzero, $\mathcal{O}$ -torsion-free, and finite over $\mathcal{A} = \mathcal{O}[[K]]$, for every sufficiently small uniform open subgroup $K \subset \mathrm{SL}_2(\mathbb{Z}_2)$. It has a continuous commuting D -action.

The construction takes place on the algebra side of the block equivalence:

$$\begin{array}{ccc} L_{\mathcal{E}^{(b)}} & \xrightarrow{-\widehat{\otimes}_{\mathcal{E}^{(b)}} P^{(b)}} & M \\ \text{forget factors } 2, \dots, b \downarrow & & \\ L_{\mathcal{E}} & \xrightarrow{-\widehat{\otimes}_{\mathcal{E}} P} & V. \end{array}$$

The subscripts indicate which algebra acts on L . The lower row reconstructs the one-factor object from that algebra module.

Here admissibility in the compact category means finite generation over the corresponding completed compact group algebra. Although K is not open in G , its product with the scalar units is open in $\mathrm{GL}_2(\mathbb{Z}_2)$; the fixed central character therefore makes this the usual admissibility condition. We prove the proposition by comparing one-factor and product coinvariants. The essential local assertion will be that, along a nonconstant characteristic-two curve, all simple block modules give the same answer to the question whether their K -coinvariants vanish.

3.1 The local block input

We record the precise consequences of Paškūnas–Tung that enter the argument. The residue field has been enlarged so that the block and its simple objects are absolutely split. Let $e \in \mathcal{E}$ be the sum of the idempotents corresponding to those simple objects on which $\mathrm{SL}_2(\mathbb{Q}_2)$ does not act trivially.

Theorem 3.2 (Paškūnas–Tung). *The following statements hold for every such block at $p = 2$, including the block of a scalar residual Galois representation.*

1. *The functor $\mathrm{Hom}(P, -)$ is an exact equivalence from the compact block to right pseudocompact $\mathcal{E}$ -modules, with inverse $-\widehat{\otimes}_{\mathcal{E}} P$. The algebra $\mathcal{E}$ is finite over R .*
2. *Let CH_R be the universal Cayley–Hamilton algebra for the local pseudorepresentation, and let $\mathrm{CH}_R^{\mathrm{tf}}$ be its maximal $\mathcal{O}$ -torsion-free quotient. There is an R -compatible identification*

$$\mathrm{CH}_R^{\mathrm{tf}} \simeq \begin{cases} (e\mathcal{E}e)^{\mathrm{op}}, & \text{in a nonsupersingular block,} \\ M_2((e\mathcal{E}e)^{\mathrm{op}}), & \text{in a supersingular block.} \end{cases}$$

The corner equivalence is induced by the Montréal functor on the quotient by the objects with trivial $\mathrm{SL}_2(\mathbb{Q}_2)$ -action.

3. *An $\mathcal{E}$ -module finite and torsion-free over $\mathcal{O}$ gives, after duality and inversion of 2, an admissible Banach representation of finite length. At an absolutely irreducible Galois coefficient parameter there is a unique absolutely irreducible Banach object. At a sum of characters $\gamma_1 + \gamma_2$, the simple objects have the following description. If the characters coincide, the unique simple is $\mathrm{Ind}_B^G(1 \otimes \varepsilon^{-1})^{\mathrm{cts}} \otimes \gamma_1 \circ \det$. If their sum is $\gamma + \gamma\varepsilon$, the simples are*

$$\{1, \widehat{\mathrm{St}}, \mathrm{Ind}_B^G(\varepsilon \otimes \varepsilon^{-1})^{\mathrm{cts}}\} \otimes \gamma \circ \det,$$

where $\widehat{\mathrm{St}}$ is continuous Steinberg. If $\gamma_1\gamma_2^{-1} \notin \{1, \varepsilon, \varepsilon^{-1}\}$, they are exactly

$$\mathrm{Ind}_B^G(\gamma_1 \otimes \gamma_2 \varepsilon^{-1})^{\mathrm{cts}}, \quad \mathrm{Ind}_B^G(\gamma_2 \otimes \gamma_1 \varepsilon^{-1})^{\mathrm{cts}}.$$

Here B is the upper Borel, induction is unnormalized, and Galois characters are viewed as characters of $\mathbb{Q}_2^\times$ by reciprocity.

The first assertion is [20, Section 4.1 and Theorems 1.2–1.3]; the second is [20, Propositions 4.10 and 4.18, Theorem 6.13]; the Banach assertions are [20, Section 4.5, Section 6.2, Equation (20), Proposition 6.9, Corollary 6.10, and Proposition 6.11]. At 2, the center comparison gives an isomorphism after inverting 2 and permits an integral cokernel killed by 2 [20, Theorem 1.4]. We use the finite map from R and the stated integral Cayley–Hamilton comparison, without identifying R with the integral center.

The compact equivalence commutes with coefficient quotients, finite presentations, and inverse limits. Consequently it preserves injections and coefficient torsion-freeness, and commutes with the completed coefficient changes below. The corresponding product equivalence is explained in [17, Proposition 4.2].

3.2 Coinvariants and the noncharacter corner

Fix a sufficiently small uniform $K \subset \mathrm{SL}_2(\mathbb{Z}_2)$, and put

$$C_K = P/(\varpi P + (K - 1)P).$$

This is a left $\mathcal{E}$ -module. Its support over R will control the admissibility test.

Lemma 3.3. *The module C_K is finite over R . For a finite pseudocompact right $\mathcal{E}$ -module W ,*

$$(W \hat{\otimes}_{\mathcal{E}} P)/(\varpi, K - 1) = W \otimes_{\mathcal{E}} C_K.$$

The analogous formula holds for product coinvariants. If a commutative complete coefficient algebra A , finite over R (over $R^{\hat{\otimes} b}$ in the product case), acts on W compatibly and commutes with its algebra action, these formulas commute with base change $A \rightarrow H$ to a field.

Proof. The object $P/\mathfrak{m}_R P$ corresponds under the compact equivalence to $\mathcal{E}/\mathfrak{m}_R \mathcal{E}$, so has finite length. Every smooth simple in the block is admissible. Its compact dual therefore has finite-dimensional K -coinvariants, with the central character understood. Right exactness of coinvariants now makes $C_K/\mathfrak{m}_R C_K$ finite-dimensional. The $\mathfrak{m}_R$ -action is topologically nilpotent: on the algebra side this follows from the finite R -module $\mathcal{E}$, and the equivalence transports the inverse limit of its finite-length quotients. Compact Nakayama proves that C_K is finite over R .

Completed tensor products commute with the cokernels defining coinvariants. After these cokernels, all modules involved are finite over their complete noetherian coefficient rings, so ordinary and completed tensor products agree. For the product formula, apply this argument in each factor. Finite presentations over R , or over its completed tensor powers, then show that the formula commutes with the asserted field base changes. $\square$

A characteristic-two curve in $\mathrm{Spec} R$ means the spectrum of a one-dimensional complete local domain quotient of R/ϖ . Its generic point is nonconstant if its map to $\mathrm{Spec} R$ is not the closed residual point.

Lemma 3.4. *At the generic point of a nonconstant characteristic-two curve, the idempotent e is full: if H is a finite extension of the curve’s fraction field and $\mathcal{E}_H = \mathcal{E} \otimes_R H$, then $\mathcal{E}_H e \mathcal{E}_H = \mathcal{E}_H$.*

Proof. We show that $\mathcal{E}/\mathcal{E}e\mathcal{E}$ is finite over $\mathcal{O}$. Under the compact equivalence its modules have only the $\mathrm{SL}_2(\mathbb{Q}_2)$ -trivial simple constituents. This is precisely the subcategory of objects with trivial $\mathrm{SL}_2(\mathbb{Q}_2)$ -action: that subcategory is thick, and the assertion passes to compact limits [20, Section 4.2].

Such an action factors through $\det : G \rightarrow \mathbb{Q}_2^\times$. On squares, the character is prescribed by the central action. Since $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$ is finite, the corresponding completed group algebra with these scalar relations is finite over $\mathcal{O}$. This remains true if the scalar relations force coefficient torsion. The regular module of $\mathcal{E}/\mathcal{E}e\mathcal{E}$ corresponds to an object of this subcategory with finite cosocle. Compact Nakayama makes it finite over that finite $\mathcal{O}$ -algebra; its endomorphism algebra, and hence $\mathcal{E}/\mathcal{E}e\mathcal{E}$, is finite over $\mathcal{O}$. Its special-fiber support over R is therefore contained in the closed point. It vanishes at the curve's generic point, as required. $\square$

3.3 The characteristic-two curve test

Let H be a finite extension of the fraction field of a nonconstant characteristic-two curve, chosen to split the parameter and the algebra $\mathcal{E}_H = \mathcal{E} \otimes_R H$. We will show that a nonzero $C_{K,H} = C_K \otimes_R H$ survives tensoring with every simple right $\mathcal{E}_H$ -module. The main case is a parameter with two distinct characters. Its two principal-series families have the same coinvariant vanishing: exchanging their inducing characters inverts their values on determinant-one stabilizers. For a unit a , the elements $a - 1$ and $a^{-1} - 1$ generate the same ideal. To apply this calculation to all block simples, we construct the families integrally and track their two character labels through specialization.

Lemma 3.5. *Suppose a nonconstant characteristic-two curve has generic local parameter $\gamma_1 + \gamma_2$, with $\gamma_1 \neq \gamma_2$. After a finite extension, let $J = k'[[t]]$ be its normalization, finite over the curve ring, and $H = \text{Frac}(J)$. The compact dual principal-series families over J with inducing orders*

$$(\gamma_1, \gamma_2 \varepsilon^{-1}), \quad (\gamma_2, \gamma_1 \varepsilon^{-1})$$

correspond to finite free J -modules W_1, W_2 for $\mathcal{E}$. Their central R -action is the given curve parameter. Their generic fibers are the two distinct simple right $\mathcal{E}_H$ -modules, and

$$(W_1 \otimes_J H) \otimes_{\mathcal{E}_H} (C_K \otimes_R H) \neq 0 \iff (W_2 \otimes_J H) \otimes_{\mathcal{E}_H} (C_K \otimes_R H) \neq 0.$$

Proof. The characters take values in $J^\times$: their values are integral by the characteristic polynomial, and their product is a unit. They are continuous. Indeed, choose an element on which the two characters differ; traces against that element recover their values continuously in H . Their reductions are the residual constituents.

We first construct the families in mixed characteristic. Enlarge the integer coefficients to $\mathcal{O}'$, and set $J_0 = \mathcal{O}'[[X, Y]]$. The pro-2 completion of $\mathbb{Q}_2^\times$ has two free generators, represented by 2 and 5, and torsion generated by -1 . Let Γ_1 be the universal lift of the first residual character on the two free generators, fixing either lift of its sign on -1 , and put $\Gamma_2 = \psi \varepsilon \Gamma_1^{-1}$. There is a continuous map $J_0 \rightarrow J$ specializing these characters to γ_1, γ_2 .

For each order, form smooth induction with coefficients in the discrete dual $J_0^\vee$, and take its compact dual N_i . This defines the family even when the inducing characters become smooth only modulo powers of the coefficient maximal ideal. Local sections on $B \backslash G = \mathbf{P}^1(\mathbb{Q}_2)$ identify its compact model with a pro-free J_0 -module. In particular the model is flat and commutes with coefficient changes. The residual smooth induction has finite length in the specified block. If the residual characters differ it is irreducible; if they coincide it has the character and Steinberg constituents. In either case it has precisely one noncharacter constituent, with multiplicity one [20, Section 4.1]. Filtering the finite coefficient quotients shows that N_i belongs to the compact block.

Put $\tilde{W}_i = \text{Hom}(P, N_i)$. Compact Nakayama and residual finite length show that this is a finite J_0 -module. Exactness of the equivalence makes it flat: for every finitely generated ideal $I \subset J_0$, apply the equivalence to the injective map $I \otimes_{J_0} N_i \rightarrow N_i$, using finite presentations to commute

these tensors with the equivalence. Thus $\widetilde{W}_i$ is finite free. The same argument proves compatibility with coefficient changes. The residual multiplicity calculation gives

$$\text{rank}_{J_0}(\widetilde{W}_i e) = 1.$$

At characteristic-zero coefficient points outside the proper loci $\Gamma_1 \Gamma_2^{-1} \in \{1, \varepsilon, \varepsilon^{-1}\}$, Schikhof duality identifies these objects with the two continuous principal series in Theorem 3.2. Those points are Zariski dense in $\text{Spec } J_0$. Hence their central R -action is everywhere the pseudorepresentation $\Gamma_1 + \Gamma_2$: equality may be tested on matrices acting on the finite free modules.

We also need to keep the two character labels distinct upon specialization. The residual block here is nonsupersingular, so Theorem 3.2 identifies the opposite corner with CH_R^{tf} , without a matrix factor. Its Galois action on each free rank-one module $\widetilde{W}_i e$ gives a character. At the dense coefficient points the two principal series are distinct simple modules, their nonzero corner modules are distinct, and their corner labels are therefore Γ_1 and Γ_2 . More explicitly, if λ_i is the corner character, set

$$I_{ij} = (\lambda_i(g) - \Gamma_j(g) : g \in G_{\mathbb{Q}_2}) \subset J_0.$$

The dense coefficient points lie in $V(I_{i1}) \cup V(I_{i2}) = V(I_{i1} I_{i2})$, so $I_{i1} I_{i2} = 0$. Since J_0 is a domain, one of the two ideals vanishes. The two labels are opposite because the generic specializations are distinct. Thus the labels on the entire family are Γ_1, Γ_2 , up to one fixed permutation, and these identities persist under $J_0 \rightarrow J$.

Set $W_i = \widetilde{W}_i \otimes_{J_0} J$. By Lemma 3.4, e is full over H ; its rank-one corner modules therefore make $W_i \otimes_J H$ simple and distinct. The specialized torsion-free Cayley–Hamilton algebra is a quotient of the Cayley–Hamilton algebra of the specialized parameter. The field structure theorem for a split multiplicity-free determinant allows at most the two character simples [5, Theorems 2.12 and 2.22]. Thus our two modules account for all simples, without assuming that forming the torsion-free quotient commutes with reduction modulo 2.

Finally calculate K -coinvariants directly in the compact induction model. There are finitely many double cosets $B \backslash G / K$. For a representative g , its contribution is J modulo the ideal generated by the inducing character minus one on $B \cap g K g^{-1}$. This follows equally by dualizing the invariant functions with values in $J^\vee$. Every stabilizer has determinant one. Since $\varepsilon = 1$ over J , the two inducing characters on that stabilizer are inverse. They generate the same ideal, because $a^{-1} - 1 = -a^{-1}(a - 1)$ for a unit a . The two coinvariant modules consequently vanish simultaneously after tensoring with H . Lemma 3.3 identifies these modules with the two displayed tensor products. $\square$

Lemma 3.6 (Uniform survival on curves). *Let H be a finite extension of the fraction field of a nonconstant characteristic-two curve in $\text{Spec } R$, large enough to split the parameter and the finite algebra $\mathcal{E}_H$. If $C_{K,H} = C_K \otimes_R H \neq 0$, then*

$$S \otimes_{\mathcal{E}_H} C_{K,H} \neq 0$$

for every simple right $\mathcal{E}_H$ -module S .

Proof. The corner is full by Lemma 3.4. The Cayley–Hamilton comparison and its field structure therefore give at most the simples labelled by the absolute constituents of the parameter. Repeated characters cannot occur on this curve. Indeed the fixed determinant has constant finite values in characteristic two; if it equals γ^2 , injectivity of squaring in a field makes γ , and hence the whole parameter, constant.

If the parameter is absolutely irreducible, the split algebra has only one simple module. A nonzero finite left module has a nonzero semisimple head, so tensoring it with that simple right

module is nonzero. If the parameter has two distinct characters, Lemma 3.5 constructs both simple modules and proves that their tensor tests have identical vanishing. At least one test is nonzero: otherwise the semisimple head of $C_{K,H}$ would be zero. Both tests are therefore nonzero. $\square$

3.4 From product coinvariants to one-factor admissibility

Proof of Proposition 3.1. The module L is finite over R , hence over the first $\mathcal{E}$. The exact compact equivalence makes V nonzero and $\mathcal{O}$ -torsion-free. It also transports the continuous commuting D -action. By Lemma 3.3,

$$Q := V/(\varpi, K - 1) = L \otimes_{\mathcal{E}} C_K$$

is finite over D/ϖ . It suffices, by compact Nakayama over $\mathcal{A}$, to show that Q is finite-dimensional over k .

Suppose otherwise. Its closed support contains a one-dimensional complete local domain quotient of D/ϖ . At the generic point of this curve, Q has nonzero fiber. Since D is finite over R , its image in $\text{Spec } R$ is also a nonconstant curve. Extend its fraction field to a field H as in Lemma 3.6. With subscripts denoting fibers, we have

$$L_H \otimes_{\mathcal{E}_H} C_{K,H} \neq 0.$$

This tensor product still has the commuting actions of the remaining $b - 1$ copies of $\mathcal{E}_H$.

For any nonzero finite right $\mathcal{E}_H$ -module U , choose a simple quotient S . Right exactness of tensor products and Lemma 3.6 give $U \otimes_{\mathcal{E}_H} C_{K,H} \neq 0$. Apply this observation successively to the remaining factors. The central parameters are equal, so the same algebra and the same module $C_{K,H}$ occur each time. We obtain

$$L_H \otimes_{\mathcal{E}_H^{b-1}} C_{K,H}^{\otimes_{\mathcal{E}_H} b} \neq 0.$$

By the product formula in Lemma 3.3, this is the fiber of $M/(\varpi, K^b - 1)$ on the chosen curve. But product admissibility makes that module finite-dimensional over k . Its continuous D -action is supported at the closed point, so its fiber on a nonconstant curve is zero. This contradiction proves that Q is finite-dimensional and hence that V is finite over $\mathcal{A}$. $\square$

4 A regular point in the trace family

The admissible module of Proposition 3.1 still need not have an algebraic vector. We first show that the size of its commuting trace family forces positive Iwasawa rank. That rank supplies algebraic vectors with a prescribed smooth type. A coefficient eigensystem among these vectors will give the regular point needed in the final tangent argument.

Keep the domain D , local block ring R , and nonzero finite D -torsion-free module L of Proposition 2.4. Thus $\dim D \geq 3$, D is finite over R , and

$$V = L \hat{\otimes}_{\mathcal{E}} P$$

is $\mathcal{O}$ -torsion-free and finite over $\mathcal{A} = \mathcal{O}[[K]]$. Here K is a sufficiently deep uniform subgroup of $\text{SL}_2(\mathbb{Z}_2)$. All coefficient fields below are finite extensions of $\mathbb{Q}_2$.

4.1 Dimension forces positive rank

Write $\mathfrak{a} = (\varpi, k - 1 : k \in K)$ for the maximal augmentation ideal of $\mathcal{A}$. We use its filtration throughout. After shrinking K , the ordered-monomial description of a uniform Iwasawa algebra gives

$$\text{gr}_{\mathfrak{a}} \mathcal{A} \simeq k[X_0, X_1, X_2, X_3]. \quad (4)$$

For $\mathbb{Z}_2$ coefficients this is [23, Theorem 3.22 and Lemmas 3.24–3.25]. The same statement for $\mathcal{O}$ follows from the ordered expansion in ϖ and $b_i = k_i - 1$, for three ordered uniform generators k_i : take K deep enough that its commutator relations have $\mathfrak{a}$ -degree at least three. The symbols of $\varpi^{a_0} b_1^{a_1} b_2^{a_2} b_3^{a_3}$ then give the polynomial basis in (4).

For a nonzero finite $\mathcal{A}$ -module M , let $g(M)$ be the degree of the eventual Hilbert polynomial

$$H_M(n) = \text{length}_{\mathcal{O}}(M/\mathfrak{a}^n M).$$

Good filtrations, the Artin–Rees property, and (4) imply $0 \leq g(M) \leq 4$. These are the filtered-algebra facts used also in the proof of [17, Proposition 4.4].

The decisive comparison is between the dimension of D and this growth degree. We will make $\dim D - 1$ successive prime specializations to a coefficient fiber chosen to have growth degree at least two: its reduction modulo ϖ is infinite-dimensional, and ϖ acts injectively. The following lemma shows that each specialization lowers growth by at least one. Thus $\dim D \geq 3$ forces the maximal growth degree four.

Lemma 4.1. *Let M be a finite $\mathcal{A}$ -module and let $f : M \rightarrow M$ be injective and $\mathcal{A}$ -linear. If $M/fM \neq 0$, then*

$$g(M/fM) \leq g(M) - 1.$$

Moreover, a finite $\mathcal{A}$ -module of growth degree four has positive rank over $\mathcal{A}$.

Proof. The induced filtration on fM is good. Artin–Rees, followed by the isomorphism $M \simeq fM$, therefore gives an integer $c \geq 1$ such that

$$fM \cap \mathfrak{a}^n M \subseteq f(\mathfrak{a}^{n-c} M) \quad (n \geq c).$$

Taking lengths in the induced exact sequence yields

$$H_{M/fM}(n) \leq H_M(n) - H_M(n - c) = O(n^{g(M)-1}).$$

This proves the first assertion, including the impossibility of a nonzero cokernel when $g(M) = 0$.

The algebra $\mathcal{A}$ is a noetherian domain and has an Ore division ring of fractions. If a finite module has rank zero, each element is annihilated by a nonzero element of $\mathcal{A}$. A filtration by its finitely many generators has cyclic torsion quotients. Such a quotient is of the form $\mathcal{A}/I$, where I contains a nonzero element a . Its associated graded module is a quotient of the polynomial ring in (4) by the nonzero initial form of a , and hence has growth degree at most three. The same bound holds for a finite extension of these quotients. Thus growth degree four implies positive rank. $\square$

Proposition 4.2. *The module V satisfies*

$$\text{rank}_{\mathcal{A}} V > 0.$$

Proof. Choose an auxiliary coefficient point y of D whose local semisimple parameter, if reducible, has character ratio outside $\{1, \varepsilon, \varepsilon^{-1}\}$. Such points exist. Indeed, at a reducible exceptional parameter the fixed determinant and the specified ratio determine each character up to a quadratic character. Local class field theory gives only finitely many quadratic characters of $\mathbb{Q}_2^\times$. The exceptional parameters therefore lie in a finite union of coefficient-point closures in $\text{Spec } R$, of integral dimension at most one. Finiteness of D over R gives the same bound for their inverse image. Since D is horizontal of dimension at least three, coefficient-point density [17, Lemma 3.3] supplies y outside that inverse image.

Put $s = \dim D$. If $\mathfrak{p}_y$ is the kernel of the coefficient point, then $D/\mathfrak{p}_y$ is finite over $\mathcal{O}$ and has dimension one. The complete local domain D is catenary, so there is a saturated chain

$$(0) = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_{s-1} = \mathfrak{p}_y.$$

Starting with $L^{(0)} = L$, define $L^{(i+1)}$ by tensoring $L^{(i)}$ with $D/\mathfrak{p}_{i+1}$ and removing its torsion over that domain. Each $L^{(i)}$ is nonzero: the preceding torsion-free finite module has full support, and Nakayama's lemma at the next prime shows that its fiber there is nonzero. Thus $L^{(i+1)}$ has nonzero generic fiber. The commuting $\mathcal{E}$ -action survives all these operations.

Choose $a_i \in \mathfrak{p}_{i+1} \setminus \mathfrak{p}_i$. Multiplication by a_i is injective on $L^{(i)}$. The exact compact block equivalence of Theorem 3.2 makes it injective on

$$V_i = L^{(i)} \hat{\otimes}_{\mathcal{E}} P,$$

and V_{i+1} is a quotient of its cokernel. All these objects are finite over $\mathcal{A}$, being successive quotients of V . Lemma 4.1 gives

$$g(V) \geq s - 1 + g(V_{s-1}). \quad (5)$$

Finally, $L^{(s-1)}$ is nonzero and finite torsion-free over $\mathcal{O}$. Its associated Banach representation has finite length and, after a finite coefficient extension, every simple factor has the central parameter of a coefficient conjugate of y ; see Theorem 3.2. They are infinite-dimensional: the generic principal-series possibilities are infinite-dimensional, as is the representation attached to an absolutely irreducible two-dimensional Galois parameter. Our choice of y excludes the exceptional character possibilities. Hence $V_{s-1}/\varpi V_{s-1}$ is infinite-dimensional over k . Otherwise compact Nakayama would make V_{s-1} finite over $\mathcal{O}$, contradicting this Banach description.

Exactness also makes multiplication by ϖ injective on V_{s-1} . The infinite-dimensional reduction has growth degree at least one, so Lemma 4.1 gives $g(V_{s-1}) \geq 2$. Together with (5), this proves

$$4 \geq g(V) \geq \dim D + 1 \geq 4.$$

The final assertion of Lemma 4.1 finishes the proof. $\square$

4.2 Capturing a fixed smooth type

Let $K_0 = \mathrm{GL}_2(\mathbb{Z}_2)$ and let

$$\sigma : K_0 \longrightarrow \mathrm{GL}_2(\mathbb{F}_2) \simeq S_3 \xrightarrow{\text{sign}} \{1, -1\}$$

be the inflated sign character over E . Its restriction to the upper unipotent subgroup of $\mathrm{GL}_2(\mathbb{F}_2)$ is nontrivial, so it is a cuspidal representation of that finite group. We use this type for two reasons. Tensoring it with an algebraic representation excludes every Banach simple with reducible Galois parameter. In a classical smooth representation it forces depth-zero supercuspidality, and hence a Weil–Deligne parameter trivial on wild inertia. The second property will exclude the CM fields inside $\mathbb{Q}(\zeta_{2^\infty})$ and thereby allow the adjoint Selmer vanishing used in Proposition 6.1.

Write

$$\Pi_V = \mathrm{Hom}_{\mathcal{O}}^{\mathrm{cts}}(V, \mathcal{O})[1/2]$$

for the unitary admissible Banach representation associated with V . Its central character on scalar units is $z \mapsto z^d$, where $d = w + 1$ by the determinant normalization. Negative determinant powers are permitted in the algebraic representations below.

Lemma 4.3. *There is an irreducible algebraic representation $U = \text{Sym}^n(E^2) \otimes \det^a$, with $n \geq 0$, $a \in \mathbb{Z}$, and $n + 2a = d$, such that*

$$\text{Hom}_{K_0}(\sigma \otimes U, \Pi_V) \neq 0.$$

This Hom space is finite-dimensional and carries a continuous commuting action of D .

Proof. Positive rank supplies a nonzero $\mathcal{A}$ -linear map $V \rightarrow \mathcal{A}$: take a nonzero linear functional after passing to the Ore division ring and clear right denominators on finitely many generators. Maps between finite $\mathcal{A}$ -modules are continuous. Choose K so that its intersection with the scalar unit subgroup Z_0 is trivial. Extend the action on the target $\mathcal{A}$ to $Z_0 K$ by the dual central character z^{-d} . Since $Z_0 K$ has finite index in K_0 , Frobenius reciprocity gives a nonzero map from V to its coinduced compact module. Dualizing gives a nonzero continuous K_0 -map

$$\mathcal{C}_d \longrightarrow \Pi_V, \quad \mathcal{C}_d = \{f \in C(K_0, E) : f(kz) = z^{-d}f(k)\}. \quad (6)$$

The action on $\mathcal{C}_d$ is left translation.

We verify the density needed to use this map. Matrix coefficients of algebraic representations of central degree zero form an algebra separating the points of K_0/Z_0 : coefficients of the adjoint representation already separate them. Polynomial approximation on compact subsets of a finite-dimensional 2-adic affine space therefore makes this algebra dense in $C(K_0/Z_0, E)$. Concretely, embed the compact quotient by the adjoint matrix entries and apply multivariable polynomial approximation on a containing compact box, after extending continuous functions by clopen partitions.

Fix one algebraic representation of central degree d , for example $\det^{d/2}$ if d is even and $E^2 \otimes \det^{(d-1)/2}$ if d is odd. Its matrix coefficients have no common zero on K_0 . On a finite clopen cover of K_0/Z_0 , divide a function in $\mathcal{C}_d$ by a nonvanishing coefficient of this representation evaluated at k^{-1} . Approximate the resulting functions of central degree zero as above, and multiply back. Complete reducibility of algebraic representations shows that the resulting dense span consists of matrix coefficients of irreducible algebraic representations with central degree d . Multiplication by the nowhere-zero function $\sigma(k^{-1})$ preserves $\mathcal{C}_d$ and its topology. Thus the matrix coefficients of the types $\sigma \otimes U$ in the statement also have dense span. A nonzero continuous map in (6) cannot kill them all.

For finiteness, transpose a type map to a continuous K -equivariant map from V to the dual of that finite-dimensional type. A finite set of $\mathcal{A}$ -generators of V determines such a map, so the space of maps is finite-dimensional. Evaluation on those generators also proves continuity of its commuting D -action. $\square$

Lemma 4.4. *Let E'/E be a finite coefficient extension, let Π be one of the E' -Banach simples described in Theorem 3.2, and let $U = \text{Sym}^n(E'^2) \otimes \det^a$, with $n \geq 0$ and $a \in \mathbb{Z}$. If*

$$\text{Hom}_{K_0}(\sigma \otimes U, \Pi) \neq 0,$$

then the Galois parameter of Π is absolutely irreducible.

Proof. Suppose the parameter is reducible. Theorem 3.2 lists its possible Banach simples. Put $S_0 = \text{SL}_2(\mathbb{Z}_2)$ and $u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. On $\sigma \otimes U$, the operator u is minus a unipotent operator. In particular $u - 1$ is invertible over E' . The group S_0 acts transitively on $\mathbb{P}^1(\mathbb{Q}_2)$. An S_0 -map from the type $\sigma \otimes U$ to a continuous principal series, followed by evaluation at the identity coset, would give a functional invariant under the upper unipotent subgroup, hence under u . This functional is zero, and

transitivity makes the whole map zero. The same argument excludes a character, whose restriction to S_0 is trivial.

For a twist of the continuous Steinberg representation, pull back the defining quotient of a continuous induction by a character along a proposed type map. This gives an extension of our finite-dimensional type by a character. Every finite-dimensional continuous representation of S_0 over E' is semisimple. To see this, continuity and logarithm charts make it analytic on a sufficiently small open subgroup. Complete reducibility for $\mathfrak{sl}_2(E')$ supplies an equivariant projection onto any invariant subspace. That projection commutes with a sufficiently small open normal subgroup of S_0 ; averaging its conjugates over the finite quotient gives an S_0 -equivariant projection. The pulled-back extension therefore splits over S_0 , producing the already excluded map into continuous induction.

This excludes every reducible parameter in the classification. $\square$

4.3 From a type eigenvector to a classical point

Proposition 4.5. *There is a coefficient point $x : D \rightarrow \mathcal{O}'$ at which the global pseudorepresentation is that of a continuous absolutely irreducible representation*

$$r_x : G_{\mathbb{Q}} \longrightarrow \mathrm{GL}_2(E'), \quad E' = \mathrm{Frac}(\mathcal{O}'),$$

after a finite coefficient extension if necessary. This representation is odd and finitely ramified. Its restriction to $G_{\mathbb{Q}_2}$ is absolutely irreducible and regular de Rham, and $\mathrm{WD}(r_x|_{G_{\mathbb{Q}_2}})$ is trivial on wild inertia. Consequently r_x is attached to a classical cuspidal eigenform up to Tate twist.

Proof. Take a type supplied by Lemma 4.3. The commuting D -action on its nonzero finite-dimensional Hom space has a common eigenvector after finite coefficient extension. It gives a continuous character $x : D \rightarrow E'$. Since D is compact, its image in the endomorphism algebra preserves a lattice; its eigenvalues are integral. Thus x takes values in $\mathcal{O}'$, and the corresponding map

$$\sigma \otimes U \longrightarrow \Pi_V[x] \tag{7}$$

is nonzero.

This eigenspace has finite length as a Banach representation. Indeed, extend coefficients to $\mathcal{O}'$, impose the scalar relations $a = x(a)$ for $a \in D$ on L , and remove $\mathcal{O}'$ -torsion. The result is finite over $\mathcal{O}'$. The exact compact equivalence identifies its inverse with the corresponding torsion-free scalar quotient of V ; continuous maps to $\mathcal{O}'$ kill the discarded torsion. Dualizing and inverting 2 therefore identifies its Banach representation with $\Pi_V[x]$. The finite-length assertion is the coefficient-fiber assertion of Theorem 3.2. A composition series now shows that some Banach simple contains the type in (7): follow a nonzero map through the series, passing to a quotient when its composite is nonzero and otherwise factoring it through the closed subrepresentation. No exactness assertion for the functor of locally algebraic vectors is involved.

Lemma 4.4 now makes the local block parameter q_x absolutely irreducible. The global determinant at x is then absolutely irreducible as well and is represented, after finite coefficient extension, by a continuous r_x ; this is the representation theorem for characteristic-zero determinants [5, Theorem 2.12]. Continuity can also be checked by recovering matrix coordinates from traces against a matrix-algebra basis. The normalization is

$$q_x = r_x|_{G_{\mathbb{Q}_2}} \otimes \varepsilon. \tag{8}$$

Since 2 splits completely in F , restriction to G_F is still absolutely irreducible.

We can now transport the type to the classical forms upstairs. The point x induces a Hecke coefficient point through $\mathbb{T} \rightarrow D$. By [17, Proposition 4.2], its completed definite-forms eigenspace contains a Banach tensor product

$$\widehat{\bigotimes_{v|2} \Pi_v},$$

where Π_v is a simple with parameter q_x . All these parameters are identical, and an absolutely irreducible parameter has a unique Banach simple. Thus each Π_v contains the type already found, and their tensor contains

$$\bigotimes_{v|2} (\sigma \otimes U).$$

These are locally algebraic vectors in the eigenspace of completed forms. The classical comparison [17, Proposition 4.1] identifies them with definite automorphic forms of algebraic factor $\bigotimes U$. The weights are regular: the factor Sym^n gives classical weight $n + 2 \geq 2$, while the determinant power only changes the common twist. In the classical decomposition the smooth factor contains σ at every dyadic place. Indeed the Lie algebra identifies the algebraic factor, and the remaining K_0 -action is its smooth multiplicity space. Reduced-norm characters cannot contain this type. Jacquet–Langlands therefore gives a regular cuspidal Hilbert eigensystem over F . Its Galois representation agrees with $r_x|_{G_F}$ by the Hecke identities and Chebotarev.

The finite-group cuspidality of σ , established above, now identifies the local smooth factors. The depth-zero type theorem identifies its characteristic-zero smooth occurrences with depth-zero supercuspidal representations [13, Appendix A, Sections A.3.1–A.3.2]. The local Langlands correspondence preserves depth [2, Theorem 2.9]; its corresponding Weil parameter therefore has depth zero, which means that it is trivial on wild inertia. The dyadic local–global comparison in [17, Proposition 4.1] now gives regular de Rhamness and this wild-inertia assertion for $r_x|_{G_F}$ at every dyadic place. The splitting of 2 in F gives exactly those properties for $r_x|_{G_{Q_2}}$. The cyclotomic twist in (8) has unramified Weil–Deligne character and does not change the wild-inertia assertion.

Finally, finite ramification is built into D , and its fixed determinant χ is odd. All the hypotheses of [17, Theorem 1.1] have now been verified for r_x : continuity, absolute irreducibility, oddness, finite ramification, and two distinct de Rham weights at 2. That theorem gives the asserted classical modularity up to Tate twist, with no restriction on the residual representation. $\square$

5 The full Hecke algebra, level, and twists

The regular point produced in Proposition 4.5 is classical up to Tate twist, and its classical level may be divisible by 2. We now show that both features are compatible with the precise Hecke algebra of Theorem 1.1. The same argument will justify the continuous cyclotomic twist used to normalize the original family.

For this section write $A(N) = \mathbb{T}_2(N)$ and $A_k(N) = \mathbb{Z}_2 \otimes_{\mathbb{Z}} T_{\leq k}^{(2)}(N)$, as in (1). Thus an integral polynomial in the Hecke generators tends to zero if it tends to zero on every fixed finite range of positive weights.

We use the following structural facts at their stated scope. The algebra $A(N)$ is reduced and is a finite product of complete Noetherian local rings with finite residue fields; the topology just defined induces the maximal-ideal topology on every factor. It carries a continuous two-dimensional determinant of $G_{\mathbb{Q}, S}$, where S consists of the primes dividing $2N$, with

$$t(\text{Frob}_{\ell}) = T_{\ell}, \quad d(\text{Frob}_{\ell}) = \ell S_{\ell} \quad (\ell \nmid 2N). \quad (9)$$

Its classical coefficient points are Zariski dense, and the displayed Hecke elements topologically generate it. These are [18, Lemmas 2.1–2.2 and Proposition 2.3]; the Frobenius convention is exactly that of Equation (2.2) there. In particular, the characteristic polynomial in this convention is $Z^2 - T_\ell Z + \ell S_\ell$. Finally, [18, Theorem 1.1 and Proposition 3.1] say, respectively, that every irreducible component has dimension four, that the closure of bounded-weight classical points has dimension at most one, and that the Eisenstein closure has dimension at most two. It follows that classical cuspidal points of weight at least three are dense on every component: remove the latter two closed loci and the other components, and apply classical density in every nonempty remaining open subset.

These facts remain valid after finite extension of integral coefficients and passage to a residue factor. Indeed this extension is finite flat. After inverting 2 it is separable, so reducedness persists; integral torsion-freeness then gives reducedness before inverting 2. Testing in all coefficient embeddings preserves the stated density. Integral finite extensions preserve component dimensions. We will use these observations when a character or an eigenpacket requires larger coefficients.

5.1 Evaluation on the ordinary tower

Proposition 5.1. *Let $N \geq 5$ be odd, and let f be a classical cuspidal eigenform of positive weight and level dividing $2^a N$, for some $a \geq 0$. Its eigenvalues away from $2N$ define a continuous homomorphism*

$$A(N) \longrightarrow \mathcal{O}_{E_f}$$

after choosing a finite 2-adic coefficient field E_f containing them. The associated determinant has the Frobenius polynomials of f in (9).

Proof. We evaluate all the forms on one ordinary tower. Density of the tame-level forms on that tower will then turn boundedness of the Hecke operators into the continuity required here.

The ordinary tower. Enlarge an integer coefficient ring $\mathcal{O}$ as necessary. Over the formal ordinary locus of the compactified tame modular curve of level $\Gamma_1(N)$, consider the tower of trivializations

$$\iota : \widehat{\mathbb{G}}_m \xrightarrow{\sim} \widehat{E}$$

of the formal group of the universal generalized elliptic curve. The ordinary locus here includes the cusps. Write $\mathcal{I}$ for the ring of integral functions on this tower, completed for the coefficient-uniformizer topology, and put $\mathcal{B} = \mathcal{I}[1/2]$, with unit ball $\mathcal{I}$. The tower is the inverse system of finite étale trivialization torsors of the ordinary connected 2-divisible group. One can see this by Cartier duality: that group's dual is étale of height one. This also proves the assertion at a multiplicative cusp. In particular, the tower and $\mathcal{I}$ are flat over $\mathcal{O}$. We use the ordinary-tower construction of [14, Section 4.2, Lemma 4.2.1 and Theorem 4.2.2]. For Drinfeld level structures we use [15, Section 3.2 and Lemma 3.5.1]; the compactified moduli interpretation, including cusps in bad characteristic, is [9, Theorem 1.2.1 and Definition 2.4.1].

A section of the i th power of the Hodge line ω evaluates to a function on this tower by expressing its pullback using the standard differential $dT/(1+T)$ on $\widehat{\mathbb{G}}_m$.

Density. We claim that the $\text{Frac}(\mathcal{O})$ -linear span of the evaluations of tame-level classical forms of positive weights is dense in $\mathcal{B}$. Only rational density is needed.

Here are the details at the prime two. Let H be the weight-one Hasse invariant on the residue curve. Its evaluation on the tower is 1, and the normalized weight-four Eisenstein series E_4 reduces to H^4 ; see [14, Section 2.1]. The open set where E_4 is nonvanishing is affine, since ω is ample. Its

formal completion is the ordinary base; denote its ring of functions by $\mathcal{I}_0$. Serre vanishing for a sufficiently high power of ω lifts H^{4m+1} to an integral form G of weight $4m+1$. Consequently

$$\eta = G/E_4^m$$

is a basis of ω on the formal ordinary base. Choose a parameter z on the formal group whose cotangent is η . Such a parameter is obtained by lifting this cotangent successively on the affine base. Express the universal trivialization as

$$\iota^* z = a_1 T + a_2 T^2 + \cdots.$$

The ring $\mathcal{I}$ is topologically generated over $\mathcal{I}_0$ by the coefficients a_j and a_1^{-1} : formal group isomorphisms are represented by their coefficients, the formal group identities, and invertibility of the first coefficient. Over the ordinary base modulo ϖ^h , the ideals cutting out the 2^n -torsion are cofinal with the powers of the parameter ideal: the formal group has height one and 2 is nilpotent on this base. Thus an isomorphism of the formal groups is a compatible system of isomorphisms of their finite flat 2^n -torsion groups. Cartier duality identifies this functor with the inverse limit of the finite étale trivialization torsors. The coefficient presentation and the tower therefore have the same coordinate ring modulo every ϖ^h . After inverting 2, formal logarithms express each individual a_j as a polynomial in a_1 with coefficients in $\mathcal{I}_0[1/2]$.

Let $\mathcal{C}$ be the closed linear span under consideration. It is an algebra, since multiplication adds positive weights. The evaluations of E_4 and G are 1 modulo the uniformizer. Their powers with exponents 2^n tend to 1, so $1 \in \mathcal{C}$; their powers with exponents $2^n - 1$ then show that their inverses also belong to $\mathcal{C}$. Ratios of equal-weight sections by powers of E_4 give all functions on the affine base, by the section-ring description of an ample line bundle. Completion gives $\mathcal{I}_0[1/2] \subseteq \mathcal{C}$. Since $\eta = G/E_4^m$ evaluates to a_1 , both a_1 and its inverse belong to $\mathcal{C}$. The logarithm identities now give every $a_j \in \mathcal{C}$, and topological generation proves $\mathcal{C} = \mathcal{B}$. No uniform bound on the denominators of all logarithm coefficients is required: each coefficient is handled separately before taking the closure.

Dyadic level and Hecke operators. We next evaluate the given form on the same tower and check that its Hecke operators preserve the integral unit ball. For forms of level $\Gamma_1(2^a N)$, choose a primitive 2^a th root of unity in the coefficient ring and transport its multiplicative torsion point by ι . This supplies the extra Drinfeld $\Gamma_1(2^a)$ -structure. The construction works on generalized elliptic curves at cusps: the original tame level is ample, and adding a subgroup in the identity component preserves ampleness. The formal group of a Tate polygon is $\widehat{\mathbb{G}}_m$ by [9, Equation (2.5.3)]. The integral sections of the Hodge line recover classical forms after inverting 2 [9, Section 4.4, Equation (4.4.2)]; hence a fixed scalar multiple of a classical form extends over the integral model and has evaluation in $\mathcal{I}$. Evaluation is nonzero for a nonzero form, as its q -expansion at a compatible multiplicative cusp shows.

The good Hecke operators preserve $\mathcal{I}$ and commute with all these evaluations. For T_ℓ the correspondence sums over degree- ℓ isogenies, with normalization ℓ^{-1} , a unit in $\mathcal{O}$. The isogenies induce isomorphisms of formal groups, and the finite flat correspondence and its trace are integral. The extension over multiplicative cusps, and the normalization of trace-pullback as ℓT_ℓ , are [9, Theorem 4.4.3 and Equation (4.5.1)]. For ℓS_ℓ , the diamond operator and the scalar change of trivialization contribute $\ell^i \ell^{-1}$ in weight i , which is the required $\ell^{i-1} \langle \ell \rangle$. This includes the diamond action on the additional 2-power level. Thus every integral polynomial in these operators is a contraction of $\mathcal{B}$.

Continuity. Let h_α be a net of such polynomials that tends to zero at each finite weight stage. On every finite sum of evaluated tame classical forms, h_α tends to zero. Density and the uniform contraction bound imply

$$h_\alpha v \longrightarrow 0 \quad (v \in \mathcal{B}) :$$

first approximate v by a finite sum, then use the contraction bound on the error. Apply this to the nonzero evaluation v_f of f . Since $h_\alpha v_f = \lambda_f(h_\alpha) v_f$, the eigenvalues tend to zero. The same argument shows that a polynomial relation in $A(N)$ acts as zero on v_f . The eigenvalue map therefore extends continuously from the dense polynomial algebra to $A(N)$. Its values are integral, either by the contraction bound or by classical integrality of good Hecke eigenvalues. Compatibility of evaluation with the Hecke action proves the asserted Frobenius identities. $\square$

5.2 Continuous cyclotomic twists

Proposition 5.2. *Let $N \geq 5$ be odd. Let $\lambda : A(N) \rightarrow \mathcal{O}_E$ be a continuous coefficient point, with associated two-dimensional determinant (t_λ, d_λ) . If E'/E is finite and $\theta : \mathbb{Z}_2^\times \rightarrow \mathcal{O}_{E'}^\times$ is continuous, then the twisted determinant*

$$((\theta \circ \varepsilon)t_\lambda, (\theta \circ \varepsilon)^2 d_\lambda)$$

is also supplied by a continuous coefficient point of $A(N)$. In particular this holds for every integer Tate twist, with the same tame level N .

Proof. Fix integral coefficients containing the residual values, and work in the factor A of the resulting Hecke algebra selected by λ . Let R^{ps} be the universal global two-dimensional determinant deformation ring for these residual data, with ramification allowed at the primes dividing $2N$ and with varying determinant. The structural facts above give a continuous surjection

$$R^{\text{ps}} \twoheadrightarrow A, \quad J = \ker(R^{\text{ps}} \rightarrow A).$$

First suppose θ has finite order, and enlarge coefficients to contain its values. Twisting the universal determinant over A defines a continuous map $R^{\text{ps}} \rightarrow A$ over these enlarged coefficients. There is no change of residual factor: every finite quotient of $\mathbb{Z}_2^\times$ is a 2-group, so a finite-order character reduces to 1 in characteristic two. At every classical cuspidal point of weight at least three, the twisted determinant belongs to a classical finite-character twist. Its extra conductor is a power of 2; Proposition 5.1 therefore supplies a coefficient point of the same $A(N)$. For $j \in J$, its image under the twisted map vanishes at all these points. Their density and reducedness show that this image is zero. Thus twisting factors through A and preserves all its coefficient points. Equality of the specialized determinants follows from (9) and Chebotarev.

Now fix λ , and fix a sign $s \in \{1, -1\}$. Every $u \in \mathbb{Z}_2^\times$ has a unique expression $u = (-1)^e 5^b$, with $e \in \{0, 1\}$ and $b \in \mathbb{Z}_2$. Define the universal character

$$\Theta_X((-1)^e 5^b) = s^e (1 + X)^b \quad \text{in } \mathcal{O}_{E'}[[X]]^\times.$$

Twisting the determinant of λ by $\Theta_X \circ \varepsilon$ gives a continuous map $R^{\text{ps}} \rightarrow \mathcal{O}_{E'}[[X]]$. For every 2-power root of unity ζ , evaluation at $X = \zeta - 1$ is a finite-order twist, so the image of every $j \in J$ vanishes at all such values. A nonzero series in $\mathcal{O}_{E'}[[X]]$ cannot do this. Indeed, after dividing by the largest common uniformizer power of its coefficients, Weierstrass preparation expresses it as a distinguished polynomial times a unit. The polynomial has only finitely many roots in the open unit disk. Therefore the image of J is zero, and the universal twisted map factors continuously through A .

For the given continuous θ , its image on the pro-2 group $1 + 4\mathbb{Z}_2$ reduces trivially in the odd-order group $k_{E'}^\times$. Hence $\theta(5) - 1$ lies in the maximal ideal of $\mathcal{O}_{E'}$. Specialize $s = \theta(-1)$ and $X = \theta(5) - 1$. This continuous specialization gives the required point. All ring topologies used here are the completed local topologies, which agree with the prescribed finite-stage topology by [18, Proposition 2.3]. $\square$

We can now pass from classical representations in the companion normalization to points of $A(N)$. If V_f denotes the arithmetic representation of a weight- k primitive form with nebentype ν , then $\det V_f = \nu \varepsilon^{k-1}$, and the wedge pairing gives

$$V_f^\vee \simeq V_f \otimes \nu^{-1} \varepsilon^{1-k} \simeq V_g \otimes \varepsilon^{1-k}, \quad (10)$$

where g is the primitive form associated with $f \otimes \nu^{-1}$. Thus an expression $r_x \simeq V_f^\vee \varepsilon^n$ becomes $r_x \simeq V_g \varepsilon^{n+1-k}$. This is the conversion in [17, Section 2.1]. Propositions 5.1 and 5.2 therefore put every representation classical up to Tate twist on $A(N)$ for a sufficiently divisible odd N .

6 From the regular point to the original representation

We have a fixed-determinant trace domain D containing both the normalized original point and the regular point x of Proposition 4.5. The latter point is now on Hecke support over $\mathbb{Q}$. To put the whole domain on that support, we compare both quotients of the global deformation ring in which the determinant is allowed to vary. The decisive fact is that this larger ring is regular at x .

6.1 The full adjoint tangent space

Proposition 6.1. *Let $r_x : G_{\mathbb{Q},S} \rightarrow \mathrm{GL}_2(E_x)$ be odd, absolutely irreducible, and classical cuspidal up to Tate twist. Suppose its restriction to $G_{\mathbb{Q}_2}$ is absolutely irreducible and de Rham with distinct Hodge–Tate weights, and its Weil–Deligne representation is trivial on wild inertia. Here S is any finite set containing 2 and all ramified primes. Then*

$$\dim_{E_x} H^1(G_{\mathbb{Q},S}, \mathrm{ad} r_x) \leq 3.$$

Consequently, if R_{glob} is the universal global determinant deformation ring with varying determinant for the semisimplified residual data of r_x , and $\mathfrak{x}$ is its coefficient prime, then

$$\mathrm{edim}(R_{\mathrm{glob}})_{\mathfrak{x}} \leq 3.$$

Both assertions permit scalar or reducible semisimplified residual data and finite extension of the coefficient ring.

Proof. Set $W = \mathrm{ad} r_x = \mathrm{End}_{E_x}(r_x)$, the full four-dimensional adjoint representation. Tameness of the Weil–Deligne parameter will give the CM qualification for adjoint Selmer vanishing; local irreducibility and distinct de Rham weights will give the local cohomology bound. For a finite place v , the finite local condition is

$$H_f^1(\mathbb{Q}_v, W) = \begin{cases} \ker(H^1(\mathbb{Q}_v, W) \rightarrow H^1(I_v, W)), & v \neq 2, \\ \ker(H^1(\mathbb{Q}_2, W) \rightarrow H^1(\mathbb{Q}_2, W \otimes_{\mathbb{Q}_2} B_{\mathrm{cris}})), & v = 2. \end{cases}$$

Here B_{cris} is Fontaine’s crystalline period ring. The global group $H_f^1(\mathbb{Q}, W)$ imposes these conditions at all finite places. Positive-degree cohomology at the real place is zero over E_x .

Newton–Thorne’s theorem [16, Theorem 5.4], specialized to $\mathrm{GL}_2/\mathbb{Q}$, states that for a regular algebraic cuspidal automorphic representation π and any prime p ,

$$H_f^1(\mathbb{Q}, \mathrm{ad} r_{\pi,p}) = 0$$

provided either π is non-CM, or its CM field K is not contained in $\mathbb{Q}(\zeta_{p^\infty})$. This statement uses the full adjoint and has no residual irreducibility hypothesis. Our classical representation is regular

algebraic, including when its classical weight is two, and twists do not change its adjoint. It remains only to verify the CM qualification.

If its CM field were contained in $\mathbb{Q}(\zeta_{2^\infty})$, the corresponding quadratic self-twist would restrict to a nontrivial wildly ramified character at 2. Indeed the three quadratic subfields are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, and $\mathbb{Q}(\sqrt{-2})$; each ramifies at 2, and a quadratic ramified character there is necessarily wild. The Weil–Deligne parameter would then be isomorphic to its twist by that character. On a wild inertia element where the character is -1 , these two parameters act as I and $-I$, respectively, by the hypothesis on x . This is impossible. Newton–Thorne therefore gives

$$H_f^1(\mathbb{Q}, W) = 0. \quad (11)$$

We next compute the local quotients. The trace pairing identifies W^* with W , so local Tate duality identifies the dual of $H^2(\mathbb{Q}_v, W)$ with $\text{Hom}_{G_{\mathbb{Q}_v}}(r_x, r_x(1))$. At 2, a nonzero such map is an isomorphism by absolute irreducibility; its determinants would force $\varepsilon^2 = 1$ on $G_{\mathbb{Q}_2}$. Thus $H^2(\mathbb{Q}_2, W) = 0$, including in weight two. At $v \neq 2$, local–global compatibility [4, Theorem A] and genericity of each local factor of a cuspidal automorphic representation give the same vanishing: the genericity criterion [1, Lemma 1.1.3] says that its Frobenius-semisimple Weil–Deligne parameter U satisfies $\text{Hom}_{\text{WD}}(U, U(1)) = 0$. A Galois morphism would induce such a morphism, also after Frobenius semisimplification. Tate twists do not affect these adjoint computations. This is the non-dyadic argument of [18, Lemma 5.2].

Write $h_v^j = \dim_{E_x} H^j(\mathbb{Q}_v, W)$. For $v \neq 2$, local Euler characteristic gives $h_v^1 = h_v^0$, whereas

$$H_f^1(\mathbb{Q}_v, W) = W^{I_v} / (\text{Frob}_v - 1)W^{I_v}$$

also has dimension h_v^0 . Hence the local quotient is zero. At 2, Schur’s lemma gives $h_2^0 = 1$, and the Euler characteristic formula gives $h_2^1 = 1 + \dim W = 5$. The filtration degrees of the filtered de Rham module $D_{\text{dR}}(W)$ are $0, 0, h, -h$ for some nonzero integer h . The Bloch–Kato dimension formula [3, Corollary 3.8.4] gives

$$\dim H_f^1(\mathbb{Q}_2, W) = h_2^0 + \dim D_{\text{dR}}(W) / \text{Fil}^0 = 1 + 1 = 2.$$

We have proved

$$\dim H^1(\mathbb{Q}_v, W) / H_f^1(\mathbb{Q}_v, W) = \begin{cases} 0, & v \neq 2, \\ 3, & v = 2. \end{cases} \quad (12)$$

The kernel of the restriction map

$$H^1(G_{\mathbb{Q}, S}, W) \longrightarrow \bigoplus_{v \in S} H^1(\mathbb{Q}_v, W) / H_f^1(\mathbb{Q}_v, W)$$

lies in $H_f^1(\mathbb{Q}, W)$, because classes in its source are already unramified outside S . Equations (11) and (12) now give the bound three.

For the embedding-dimension assertion, apply the characteristic-zero tangent comparison of [18, Proposition 4.1] to R_{glob} . Its trace and determinant values topologically generate the ring, and the specialization at $\mathfrak{x}$ is absolutely irreducible. The comparison bounds the embedding dimension by $\dim_{E_x} H^1(G_{\mathbb{Q}, S}, W)$, with no condition on the semisimplified residue. The continuity needed here follows because $R_{\text{glob}}/\mathfrak{x}^2$ is finite over $\mathcal{O}$: $R_{\text{glob}}/\mathfrak{x}$ is finite over $\mathcal{O}$, and $\mathfrak{x}/\mathfrak{x}^2$ is finite over that quotient. Finite extension of $\mathcal{O}$ adds no tangent directions, since characteristic-zero derivations kill its fraction field by separability over $\mathbb{Q}_2$. The preceding cohomological bound therefore gives the asserted embedding dimension. $\square$

6.2 A regular point determines the supporting component

The following elementary lemma isolates the use of the four-dimensional Hecke theorem. It explains why the tangent calculation is needed in addition to the existence of a modular point.

Lemma 6.2. *Let R_0 be a complete Noetherian local $\mathcal{O}$ -algebra, and let A_0 and D_0 be quotients of R_0 , with D_0 a domain. Suppose every irreducible component of $\mathrm{Spec} A_0$ has dimension four. Let $x_0 : R_0 \rightarrow \mathcal{O}'$ be a continuous coefficient point factoring through both quotients, where $\mathcal{O}'/\mathcal{O}$ is a finite extension of integer coefficient rings. If*

$$\mathrm{edim}(R_0)_{\mathfrak{x}_0} \leq 3, \quad \mathfrak{x}_0 = \ker x_0,$$

then $R_0 \rightarrow D_0$ factors through A_0 .

Proof. Choose a minimal prime P of A_0 contained in its prime $\mathfrak{x}_0$. The complete local domain A_0/P is catenary. Moreover $A_0/\mathfrak{x}_0$ is an $\mathcal{O}$ -submodule of $\mathcal{O}'$, hence finite over $\mathcal{O}$, and it contains $\mathcal{O}$. It therefore has dimension one. The dimension formula gives

$$\dim(A_0/P)_{\mathfrak{x}_0} = \dim A_0/P - \dim A_0/\mathfrak{x}_0 = 4 - 1 = 3.$$

As R_0 surjects onto A_0 , we obtain

$$3 \leq \dim(A_0)_{\mathfrak{x}_0} \leq \dim(R_0)_{\mathfrak{x}_0} \leq \mathrm{edim}(R_0)_{\mathfrak{x}_0} \leq 3.$$

Thus $(R_0)_{\mathfrak{x}_0}$ is a regular local ring of dimension three, in particular a domain. Its quotient $(A_0)_{\mathfrak{x}_0}$ has the same dimension, so the localized kernel $\ker(R_0 \rightarrow A_0)_{\mathfrak{x}_0}$ is zero: a nonzero ideal in this regular local domain has quotient of smaller dimension.

If $a \in \ker(R_0 \rightarrow A_0)$, there is consequently $s \notin \mathfrak{x}_0$ with $sa = 0$ in R_0 . The kernel of $R_0 \rightarrow D_0$ is contained in $\mathfrak{x}_0$, so the image of s in D_0 is nonzero. Since D_0 is a domain, the image of a must vanish. This proves the factorization. $\square$

Proof of Theorem 1.1. Apply Proposition 2.4 to the original representation, after its indicated continuous cyclotomic normalization and finite coefficient extension. It supplies the horizontal trace domain D with the normalized original coefficient point. Propositions 3.1, 4.2, and 4.5 supply a coefficient point x of D satisfying all the hypotheses of Proposition 6.1.

Choose an odd $N \geq 5$ sufficiently divisible that it contains every odd ramified prime of the original family and that the classical representation underlying r_x has level dividing $2^a N$ for some a . Increase the allowed global ramification set to the primes dividing $2N$; the existing family remains unramified at any newly added primes. By Propositions 5.1 and 5.2, r_x gives a continuous point of $\mathbb{T}_2(N)$ with the required Frobenius values.

Fix the coefficient ring $\mathcal{O}$ of D and the residual embedding selected by x . Let A be the corresponding residue factor of $\mathcal{O} \otimes_{\mathbb{Z}_2} \mathbb{T}_2(N)$, and let R_{glob} be the universal global determinant deformation ring for this residual determinant with *varying* determinant. Let $\mathcal{O}_x$ be a coefficient ring for x . The determinant on D and the Hecke determinant give a commutative diagram whose top arrows are continuous surjections:

$$\begin{array}{ccc} & R_{\mathrm{glob}} & \\ \swarrow & & \searrow \\ D & & A \\ \searrow & & \swarrow \\ & \mathcal{O}_x & \end{array}$$

The lower arrows are the coefficient point x ; equality follows from its Frobenius identities and Chebotarev. The quotient D has fixed determinant χ , whereas R_{glob} and A retain determinant variation. Surjectivity onto A follows from topological generation by the Hecke elements and compactness, as in [18, Proposition 2.3]; surjectivity onto D is its definition as a trace-image quotient.

Every component of A has dimension four by [18, Theorem 1.1] and finite coefficient extension. Proposition 6.1 gives $\text{edim}(R_{\text{glob}})_x \leq 3$. Lemma 6.2 therefore makes $R_{\text{glob}} \rightarrow D$ factor through A . The normalized original coefficient point of D is consequently a point of the required Hecke algebra over $\mathbb{Q}$.

Undo the normalization with Proposition 5.2. We obtain, over a finite extension E'/E , a continuous map

$$\lambda : \mathbb{T}_2(N) \longrightarrow \mathcal{O}_{E'}$$

whose values on T_ℓ and ℓS_ℓ are respectively $\text{tr } r(\text{Frob}_\ell)$ and $\det r(\text{Frob}_\ell)$ for $\ell \nmid 2N$. The compact image of r preserves an $\mathcal{O}_E$ -lattice, so all these values lie in $\mathcal{O}_E$. Since the Hecke generators topologically generate $\mathbb{T}_2(N)$ and $\mathcal{O}_E$ is closed in $\mathcal{O}_{E'}$, the entire image of λ lies in $\mathcal{O}_E$. The induced map to $\mathcal{O}_E$ is continuous for its 2-adic topology. Our choice of N includes every odd ramified prime of the original r , which completes the proof. $\square$

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